BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 258 · Processor and memory synoptic challenge

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is an ADC used for?

Q2. Which register holds the current instruction?

Q3. What is the main limitation of two cores on wholly dependent steps?

Q4. A full adder receives 1,1 and carry-in 0. Give sum and carry-out and explain.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

PC — Program Counter; next instruction address.

MAR — Memory Address Register.

MDR — Memory Data Register.

cache — Fast nearby memory.

throughput — Work completed per unit time.

New learning

Fetch the instruction — PC identifies address 72, which MAR sends to memory. MDR transfers its contents to CIR.

Differentiate performance improvements — Cores can execute independent work; cache helps when needed data/instructions are present.

Explain limits — Branching, shared memory bandwidth, serial dependencies and software constrain real speedup.

Worked example

Starting PC=40, trace MAR, CIR and PC after a sequential fetch.

1. Predict — Copy PC=40 into MAR to select the instruction location.

2. Explain — Instruction contents travel via MDR to CIR, while sequential PC advances to 41.

We Do · Practise with guidance

Two independent jobs take 5 and 3 minutes. Calculate ideal two-core completion time.

```
Sequential: 5+3 = 8
Concurrent: max(5,3) = 5
```

Predict

Each available core receives a separate independent job.

Explain

Ideal duration is the longer job, five minutes; serial execution would take eight.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State what PC, MAR, MDR, CIR and ACC contain or do.

Question 2

Explain how cache and extra cores could help video rendering, and why speedup may be less than expected.

Question 3

Compare a multicore CPU and GPU for an interactive logic controller and a million independent pixels.

Exit Ticket · Answer before checking

Doubling CPU cores always halves runtime.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Converting an analogue sensor signal into a digital representation.

Q2. CIR.

Q3. The next step cannot start until the earlier result exists.

Q4. $1+1+0=10_2$; sum bit 0, carry-out bit 1.

You Do pages · Independent questions

Q1. [5] PC next instruction address [1], MAR memory address [1], MDR transferred data/instruction [1], CIR current instruction [1], ACC interim arithmetic result [1].

Q2. [6] Cache can reduce some slow main-memory accesses [2]; cores can process independent frames or tiles [2]; dependencies, bandwidth and overhead reduce ideal gains [2].

Q3. [6] General control with branching suits CPU [2]; similar pixel calculations suit many GPU lanes [2]; data transfer and parallelisation overhead determine actual suitability [2].

Exit Ticket · Model answer

False. Only suitably parallel work can benefit and overhead limits scaling.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding